

Image Processing of Die Attach's X-Ray Images for Automatic Voids Detection and Evaluation

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Abstract: The goal of this paper is to present the method for automatic detection and location of power chip, which is soldered on DBC substrate, and voids in die attach solder joint. The method of chip location is provided by two dimension correlation applied on X-Ray images. When chip location is detected, voids in solder joints can be detected. It is very important to make such analysis, because during soldering process lot of voids and defects in die attach can occur. Basic indicator of joints' quality is quantity and size of voids in die attach. Quality of solder joints depends on homogeneity of solder, which have the crucial effect on it. Automatic detection of chip and voids in die attach is achieved by techniques of image processing like filtering, energetic normalization, thresholding and image enhancement. In the paper, results of analyses of more solder alloy (99.9Sn, SnCu3In0.5, 96Sn4Ag, 99Sn1Ag) are presented. This paper offers the effective and precise method for automatic detection and evaluation of the voids from X-Ray images of solder joint.

1. INTRODUCTION

Die attach in power electronic are the subject of interest because they have to fulfill the requirements e. g. high thermal conductivity, mechanical strength, electrical conductivity and high reliability. These properties are most affected by voids in die attach [1]. In many cases, voids have large dimensions or they can appear in higher quantity what caused the degradation of joints. Even a small void at the wrong location under the chip could be critical. Therefore, location of the voids should be recognized by the process engineer to assure a reliable die attach. X-Ray analysis is one of the possibility for voids detection in die attach. The quality of solder joints could be evaluated by process engineers only using the visual control of X-Ray images yet. The article presents the automatic voids' detection in X-Ray images of die attach. The quality evaluation of die attach by process engineers could be simplified through the described method, which allows the automatic evaluation of voids' size and quantity due to the image processing [2]. For this reason, the method presented in the paper makes a significant contribution to improving the evaluation of solder joints' quality. The results would be used to analyze the impact of voids on thermal conductivity as well as the mechanical strength of die attach. In the paper we suggest the algorithms of

X-Ray images processing implemented by Matlab. In the chapter two techniques of image processing are analyzed. The chip has not always the same position on substrate. Hence, it is necessary to localize the area of interest. During the process of analysis, 2D correlation is used. Non-homogenous exposition of samples during X-Ray imaging is the common cause of errors in automatic detection of voids or other imperfections. In the article a method of progressive processing that is based on segmentation and energy normalization is proposed. In the chapter three results of detection are evaluated.

2. IMAGE PROCESSING OF X-RAY IMAGES

The issue of the X-Ray images analysis and processing is well known not only in medicine but also in area of industry production. The processing of X-Ray images in the field of industry is essential for evaluation of quality and reliability of many products [3, 4, 5]. For this intention, the image processing techniques are used. The paper is focused on evaluation of quality of die attach under the chip. The quality is mainly given by the homogeneity of the alloy [6]. In Fig. 1 is shown block diagram of the proposed system for detection and evaluation of inhomogeneity caused by the voids.

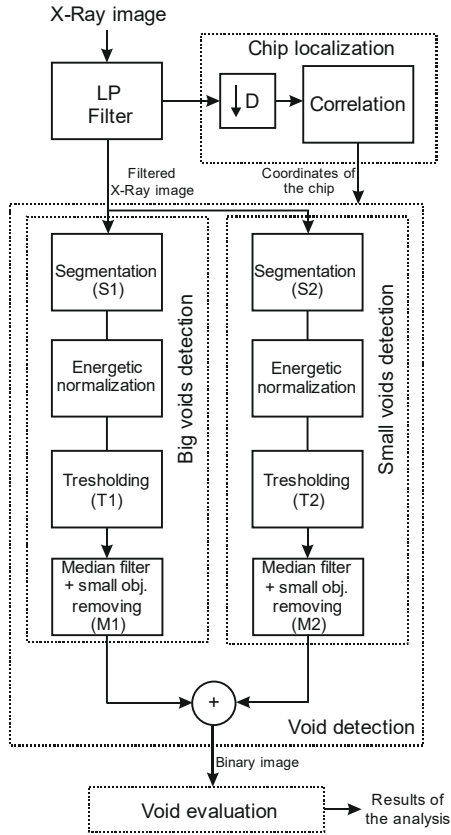


Fig. 1. Block diagram of the system for automatic voids detection and evaluation.

It is possible to see in Fig. 1 that proposed system consist of four essential blocks. The image on input is at first filtered by the low-pass discrete filter [7]. By this filter the noise suppression is achieved. For needs of automatic detection, the filtered image is decimated by decimator with decimation factor D . In the next step, the location of the chip on substrate is determined. The chips' coordinates and the filtered image are brought on input of the detection block. The analysis is performed in two parallel branches. In the process of image capturing, samples are not always uniformly exposed to radiation of the X-Ray, so the consequential images is not possible normalize as a whole. From the given reason the image has to be segmented into an orthogonal blocks. The segmentation removes issues arising from non-uniformly exposition of image, but segmentation lead to complication in detection of voids. The big blocks (over 64×64 pixels) are suitable for big voids detection. For detection of small voids it is better use a block with smaller size. The algorithms of voids detection are in other way the same. Output of detection is binary image, which represents detected voids. So, then the binary image is statistically

analyzed. In the next subchapters the previously mentioned techniques are described in detail.

2.1. Localization of die attach

Determination of chip's position has to be done before voids detection and evaluation in die attach. The chip's placement it is possible to determine manually on the base of subjective technologist's decision or it may be determined by automatic. In the cases of more samples, it is better if position is determined by automatic software solution. The position of arbitrary component can be determined by using of correlation [8]. It is achieved on the base of correlation between decimated image and assumed shape of chip. Calculation of the correlation is a time consuming process which has high requirements on the hardware. For this reason, the input X-Ray image is decimated by the factor D . By the experiments it has been observed that the factor 4 is, with regards on speed and accuracy of computation, optimal. Decimation by the factor 4 leads to the 16 times faster calculation. Value of the correlation in the point $r_{u,v}$ is given by eq. (1).

$$r_{u,v} = \frac{\sum_i \sum_j (X_{u+i,v+j} - \mu_X)(Y_{i,j} - \mu_Y)}{\sqrt{(\sum_i \sum_j (X_{u+i,v+j} - \mu_X))^2 (\sum_i \sum_j (Y_{i,j} - \mu_Y))^2}} \quad (1)$$

where X is analyzed image, Y is image of required shape and μ_X a μ_Y are average values of these images.

The point $r(u, v)$ with maximal correlation's value represents the position of left upper corner of the chip. Obtained coordinates has to be multiplied by value of decimation factor, because analysis has been performed in decimated image. The 3D graph of the correlation function is shown in Fig. 2. The Fig. 3 shows the X-Ray image of substrate and the alloy under the chip. The automatic detection of die attach is highlighted by red square.

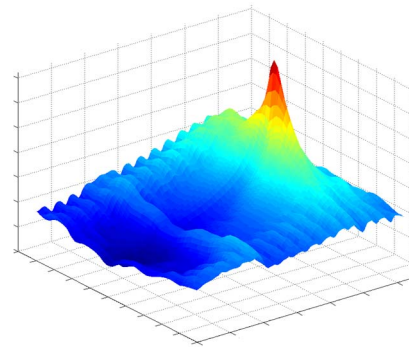


Fig. 2. 3D graph of the correlation function.

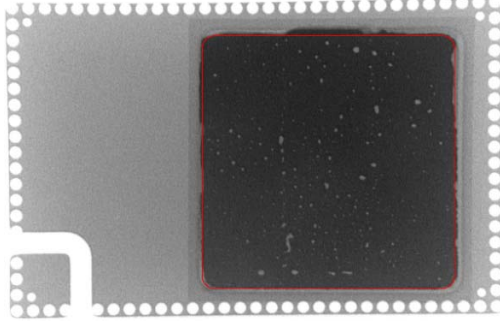


Fig. 3. X-Ray image of die attach and detected die attach.

2.2. Segmentation

As it is mentioned earlier, images which were captured by X-Ray in not ideal conditions of radiation, is not possible normalize as a whole. In Fig. 4(a) the typical example of non-uniformly exposed image is shown. Fig. 4(b) shows energetically normalized image and in Fig. 4(c) is this normalized image by thresholding converted into its binary form.

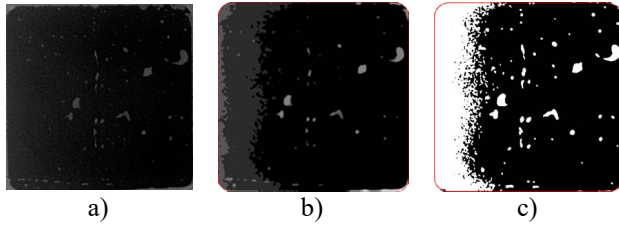


Fig. 4. a) X-Ray image of non-uniformly exposed die attach, b) this image after energetic normalization as a whole and its c) binary image.

It is clear, that binary image from Fig. 4(c) cannot be used for evaluation of multitude and arrangement of voids. By segmentation of image on to orthogonal blocks et seq. by the energy normalization inside each block it is possible to eliminate this issue. In this manner, normalized and quantized image from Fig. 4(a) for blocks' size 16x16 pixels is shown in Fig. 5(a) and Fig. 5(b) shows this image for blocks' size 64x64 pixels. Finally in Fig. 6(c) is shown the binary image achieved by merging of binary images given from thresholded images

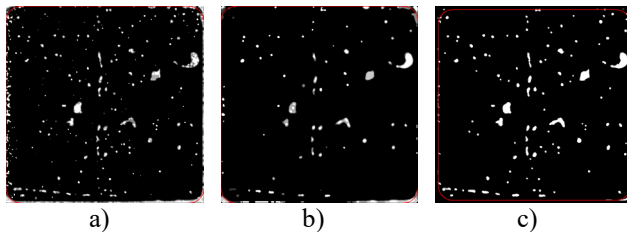


Fig. 5. The energetically normalized X-Ray image of non-uniformly exposed die attach processed by blocks with size a) 16x16 pix, b) 64x64 pix and c) combined binary image.

2.3. Energetic normalization

Images can be differently exposed in regards to conditions in which they were captured. Consequently before the next analysis, images have to be normalized [9]. By the energetic normalization all analyzed images gets into the same energetic range. From previous chapters arises that normalization is better to perform by the blocks. The normalization is achieved by applying eq. (2) on each block, with size $M \times N$ pixels, of image.

$$\tilde{x}_{u,v} = \left\lfloor \frac{x_{u,v} - E(x)}{\sigma(x)} \right\rfloor \quad (2)$$

where $\tilde{x}_{u,v}$ is energetic normalized value of a pixel $x_{u,v}$. $E(x)$ and $\sigma(x)$ represent average value and dispersion of analyzed block. Average value and dispersion from eq. (2) are given by eq. (3) and (4).

$$E(x) = \frac{1}{NM} \sum_{u=1}^N \sum_{v=1}^M x_{u,v} \quad (3)$$

$$\sigma(x) = \sqrt{\frac{1}{NM} \sum_{u=1}^N \sum_{v=1}^M (x_{u,v} - E(x))^2} \quad (4)$$

By the experiment it has been concluded that $E(x)$ from numerator of eq. (2) is advisable determine on the base of eq. (3) applied on the blocks of image. But in the other side denominator of eq. (2) is better determined by eq. (4) applied on image as a whole. Furthermore, it is clear that pixels, which have negative value after normalization can be replaced by zero.

2.4. Binarization

The conversion of energetically normalized image into a binary form, also called thresholding, is the last step of voids detection [10]. By thresholding binary value assignment to pixel is achieved. If value of the pixel x_p exceed the value of threshold t , the binary pixel x_b on the position of x_p get binary value 1. In other cases the binary pixel x_b has value of logical zero. This is expressed by eq. (5).

$$x_b = \begin{cases} 0, & x_p < t \\ 1, & x_p \geq t \end{cases} \quad (5)$$

Furthermore, the binary image is filtered by a median filter [11]. This filtration is involved for suppression of pulse noise. Also, small objects are removed from the

binary image. Small objects represent wrong detection or voids with insignificant size.

2.5. Voids evaluation

The voids in die attach have significant influence on joint's quality. For this reason it is necessary to evaluate size, number and deposition of voids in solder joints. A joint are characterized as suitable when total solder porosity is lower than 15% of total die surface and a size of lager voids not exceed 3% of total die surface [3]. The lager voids or lot of small voids can decrease electrical, mechanical and thermal properties of die attach [4]. Even a small void under the chip could has significant influence on thermal conductivity of die attach [6]. It is important to determine number and average value of void's size. For evaluation of the void's influence on the quality of die attach, voids are divided in to these categories: 0.05 – 0.5%, 0.5 – 1%, 1 – 3% and more than 3% of total die surface. Voids smaller than 0.05% have insignificant influence on the quality of die attach but contributions of all voids cannot be above 15% of total die surface.

3. RESULTS

The techniques mentioned in previous chapters were implemented in program suite Matlab. Fig. 6 shows X-Ray images of analyzed die attach, where a green square represents the result of automatic join's detection and detected voids are bordered by red. In Fig. 6(a, b, c) die attach with solder consist of alloy 96Sn4Ag are shown. In Fig. 6(d, e, f) die attach consists of 99.9Sn are shown. Die attach created by alloy of 99Sn1Ag in Fig. 6(g, h, i) and SnCu3In0.5 in Fig. 6(j, k, l) are shown. Hence, it is clear that images were acquired in different conditions and settings of the X-Ray device. From figures on Fig. 6 it is possible to see, that experiments are performed in a different size and distribution of voids. Die attach for individual alloys are specified as S_a , S_b , ..., S_l . In the meaning of subchapter 2.5, voids of individual die attach are evaluated in Table 1. In the Table 1 there are listed statistical properties of detected voids like total perceptual value of voids in alloy, maximal value, average value, deviation and number of voids with defined size. As it is presented in Table 1, samples S_a and S_b are unsuitable, because in these samples voids occupy more than 3% of total die surface. Samples S_d , S_e and S_f are on the limit of requirements. According to results of the analysis, all of the other samples fulfilled all requirements mentioned above.

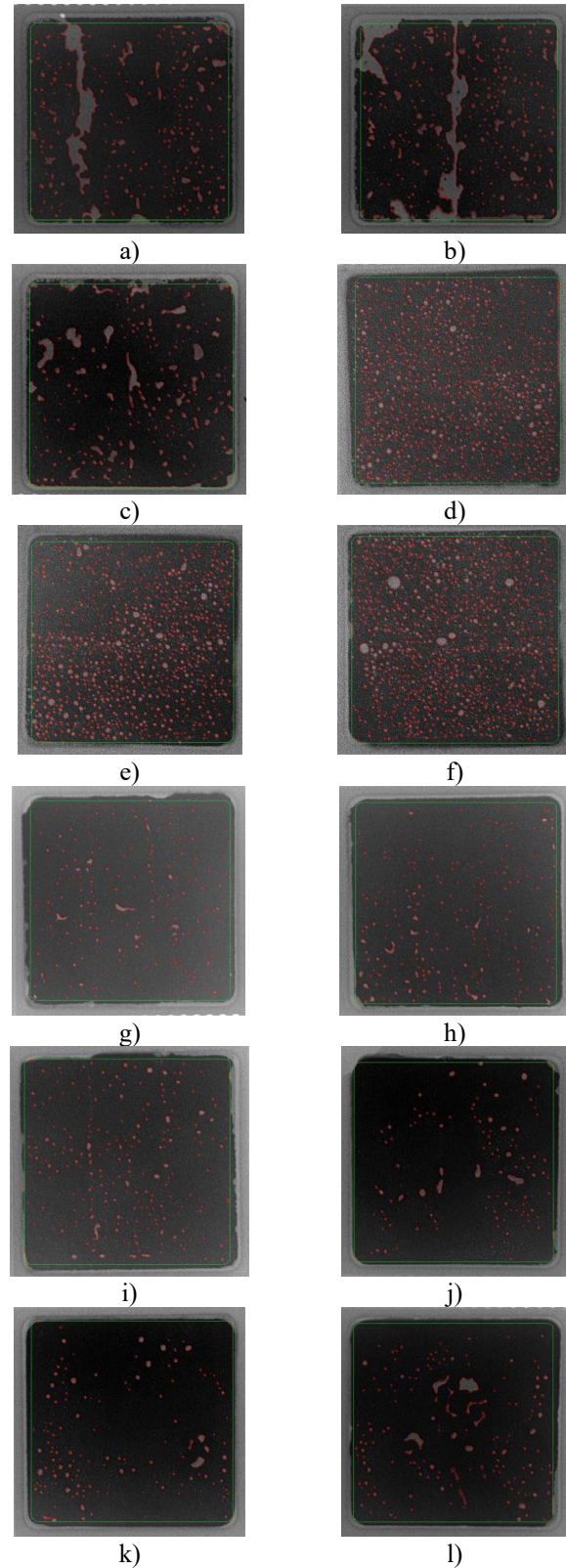


Fig. 6. X-Ray images with marked voids in Die attach with solder consist of a, b, c) 96Sn4Ag; d, e, f) 99.9Sn; g, h, i) 99Sn1Ag; j, k, l) SnCu3In0.5.

Table 1. Statistical evaluation of voids in tested die attach

Samples		Global statistical parameters of voids					Number of voids with size				
Label	Alloy	Area [%]	Avg. value [%]	Dev. [%]	Max. size [%]	Total number	below 0.05%	between 0.05% and 0.5%	between 0.5% and 1%	between 1% and 3%	Over 3%
S _a	96Sn4Ag	8.941	0.045	0.311	4.353	197	175	21	0	0	1
S _b		12.632	0.065	0.417	5.375	194	174	18	1	1	1
S _c		7.346	0.042	0.084	0.625	174	146	26	2	0	0
S _d	99.9Sn	11.812	0.014	0.010	0.110	859	846	13	0	0	0
S _e		10.715	0.015	0.012	0.097	693	675	18	0	0	0
S _f		11.042	0.014	0.015	0.283	772	762	11	0	0	0
S _g	99Sn1Ag	2.287	0.012	0.011	0.111	194	192	2	0	0	0
S _h		2.331	0.013	0.012	0.094	174	170	4	0	0	0
S _i		2.689	0.013	0.011	0.113	209	207	2	0	0	0
S _j	SnCu3In0.5	1.852	0.017	0.024	0.186	106	101	5	0	0	0
S _k		2.069	0.016	0.019	0.176	127	121	6	0	0	0
S _l		3.120	0.020	0.045	0.465	158	151	7	0	0	0

4. CONCLUSION

In the paper the techniques of image processing applied in the field of detection and evaluation of voids in die attach are analyzed. The system of automatic localization of die attach under the chip has been proposed. After that the two possible approaches to the energetic normalization of X-Ray images are analyzed. On the base of this analysis the method of segmented X-Ray images normalization has been proposed. Normalized X-Ray images obtained in such a way are turned into a binary form. These binary images represents detection of the voids. By the statistical analysis a quality evaluation of die attach is achieved.

5. ACKNOWLEDGEMENT



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